

Course Syllabus
EGCI 230 Electric Circuit Analysis

- 1. Program of Study** Bachelor of Engineering Program in Computer Engineering (International Program)
- 2. Course Code/Title** EGCI 230 Electric Circuit Analysis
- 3. Number of Credits** 4 (4-0-8) Credit (Lecture-Lab-Research)
- 4. Prerequisites** General Physics
- 5. Type of Course** Major Course (Required Major) (Room : 1403 Wed. 14:00-16:00 and Thru. 8:00-10:00)

Facebook : EGCI 230 ECA_QII_2026
<https://www.facebook.com/groups/1543320893628796>

- 6. Session / Academic year**
This course will be offered every the 2nd or 3rd trimester starting from the academic year since 2016.

- 7. Course Conditions** Class size will be in the range of 5-40 students.

8. Course Description

Circuit elements; Kirchhoff's laws and reference direction; elementary concepts of network graphs; resistive circuits; node and mesh analysis; Thevenin theorem and Norton theorem; Op-amp, inductor and capacitor, first-order and second order circuit; step responses; zero input and zero-state responses; DC transient and AC steady-state response; elementary transfer functions; sinusoidal waveforms; phasor representations; impedance and admittance; sinusoidal steady-state analysis; complex power; frequency response

9. Course Objectives

After successful completion of this course, students will be able to

- 9.1 Gain an in-depth understanding of theories related to the electric circuits.
- 9.2 Understand steps and procedure of circuit analysis.
- 9.3 Providing the basic knowledge for the consecutive courses such as electronics, digital circuits and IC design.

10. Course Outline

week	Topics	Hours			Teaching methods/multi media	Instructor
		Lecture	Lab	Self-Study		
1	Introduction	4	0	8	Lecture	Asst. Prof. Decha Wilairat
2	Kirchhoff's Law: KCL and KVL. Basic Resistor Circuits	4	0	8	Lecture, exercises and case studies	
3- 4	Nodal and Mesh circuit analysis.	8	0	16	Lecture and exercises	
5	Thevenin and Norton theorem	4	0	8	Lecture, exercises and group discussion	
6	Op-Amp	4	0	8	Lecture, group assignment and case studies	

week	Topics	Hours			Teaching methods/multi media	Instructor
		Lecture	Lab	Self-Study		
7	Capacitor and Inductor	4	0	8	Lecture and exercises	Asst. Prof. DechaWilairat
	Midterm Examination				TBA (Probably Wed.11 Feb.)	
8	Response of RC 1 st order circuit	4	0	8	Lecture and exercises	
9	Response of RL 1 st order circuit	4	0	8	Lecture and exercises	
10	AC steady-state response	4	0	8	Lecture and exercises	
11	AC Phasor circuits	4	0	8	Lecture and exercises	
12	Complex power	4	0	8	Lecture, exercises	
13	Final Examination				Thru. 2nd April 8:00 -10:00	
	Total	48	0	96		

11. Teaching Method

Lecture, group discussion, and oral presentation.

12. Teaching Media

Lecture handouts, PowerPoint, multimedia, CAI, etc.

13. Measurement and Evaluation of Student Achievement

Evaluate student's achievement from:

- 13.1 Ability to apply the knowledge to solving engineering problems.
- 13.2 Ability to analyze engineering problem systematically.

Student's achievement will be evaluated according to the faculty and university standard, using the symbols: A, B, B+, C, C+, C, D+, D and F.

Weight:

1. Midterm exam	40 %
2. Final exam	40 %
3. Homework	10 %
4. Quizzes and class attendance	10 %
Total	100 %

Grading scale	Grade	Point
90 - 100	A	4.0
85 - 89	B+	3.5
80 - 84	B	3.0
75 - 79	C+	2.5
70 - 74	C	2.0
65 - 69	D+	1.5
60 - 64	D	1.0
Below 60	F	0

Notes for more grading policy :

- a) For those students who receive the total score below 60% will get a grade "F" and above 90% will be an "A". The other grades depend on an average of the class like the tentative scale above.

- b) Both exams are close book, no calculator. But you are allowed to use the “formula sheet” (size A4 – one side) attached with the exam paper during the test.
- c) Please come on time to the lecture. I will randomly check your attendance. You are able to absent the lectures three times. Your grade will be deducted 1% for each lecture you missing the class (if you have more than three times). There are no make-up the labs/quizzes in classes for the students who miss them.

14. Course Evaluation

- 14.1 Evaluate as indicated in number 13 above.
- 14.2 Evaluate student’s satisfaction towards teaching and learning of the course using a questionnaire.

15. References

William H. Hayt Jr., Jack E. Kemmerly and Steven M. Durbin “Engineering Circuit Analysis”, 8th Edition, McGraw Hill 2013

16. Instructors

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17. Course Coordinator

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